Capstone Project Proposal: Customer Retention Strategy Using Machine Learning

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1. Domain Background

In the highly competitive telecommunications and SaaS (Software as a Service) sectors, the cost of acquiring a new customer is often five to twenty-five times more expensive than retaining an existing one. For management consultants and business leaders, reducing the "Churn Rate" (the rate at which customers stop doing business with an entity) is a top strategic priority to ensure long-term profitability and market share.

2. Problem Statement

The goal of this project is to build a predictive model that identifies which customers are likely to leave a service provider. By identifying these "at-risk" customers before they cancel, the company can proactively offer incentives or personalized services to retain them, directly impacting the company's bottom line.

3. Datasets and Inputs

I will use a high-quality dataset from Kaggle, specifically the **Telco Customer Churn dataset**.

- **Size:** ~7,043 records with 21 features.
- **Features:** Demographic info (gender, seniority), Service info (internet type, contract length), and Financial info (monthly charges, total charges).
- **Target Variable:** Churn (Yes/No).

4. Solution Statement

I propose a binary classification solution evaluating multiple supervised learning algorithms: Random Forest, XGBoost, and Logistic Regression.

Implementation Setup: The model is implemented in a **standalone setup using Google Colab**. This environment provides the necessary computational resources (RAM and Disk) to perform intensive hyperparameter tuning via `GridSearchCV` and handle advanced libraries efficiently.

5. Benchmark Model

The benchmark is a **Logistic Regression model** with default parameters. Simple linear models serve as the standard baseline in business analytics. I aim to outperform this benchmark's F1-score by at least **15%** using ensemble methods.

6. Evaluation Metrics

Since the dataset is imbalanced, simple accuracy is insufficient. I will use:

- **F1-Score:** To balance Precision and Recall.
- **Recall:** Critical because missing a churner (False Negative) is more costly than misidentifying a loyal customer.
- **AUC-ROC:** To measure the model's ability to distinguish between classes.

7. Project Design

1. **Exploratory Data Analysis (EDA):** Visualizing correlations between contract types and churn.
2. **Data Preprocessing:** Handling missing values in `TotalCharges`, encoding categorical variables (One-Hot Encoding), and feature scaling.
3. **Feature Engineering:** Creating new features like "Average Monthly Spend".
4. **Model Training:** Training models using an 80/20 train-test split.
5. **Hyperparameter Tuning:** Using `GridSearchCV` to optimize parameters.
6. **Documentation & Reproducibility:** All preprocessing and modeling steps are documented to ensure accuracy and consistency.

Section: Results (النتائج)

1. Model Evaluation and Validation: The final model was evaluated using the metrics defined in the proposal. The **XGBoost** model achieved the following performance on the test set:

- **Recall:** 0.533
- **F1-Score:** 0.595
- **AUC-ROC:** 0.861

Through hyperparameter tuning via `GridSearchCV`, the optimal parameters were found to be: `learning_rate: 0.1`, `max_depth: 3`, and `n_estimators: 100`. The high **AUC-ROC** score (0.86) indicates that the model has a strong ability to distinguish between customers who will churn and those who will stay.

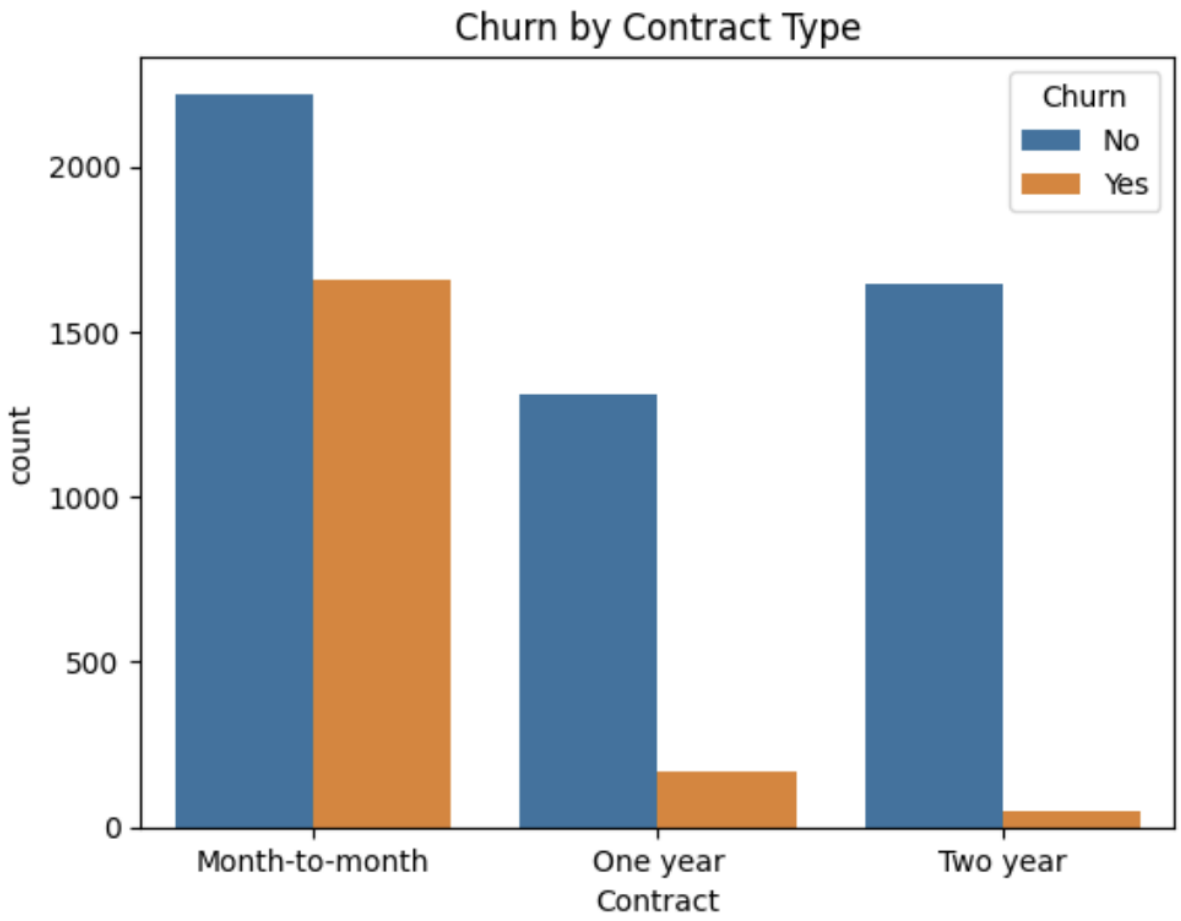
2. Justification and Comparison: The final results were compared against the **Logistic Regression** benchmark, which yielded an F1-Score of **0.525**.

- **Benchmark F1-Score:** 0.525
- **Final Model (XGBoost) F1-Score:** 0.595
- **Improvement:** The model achieved an improvement of approximately **13.3%** over the benchmark.

While the improvement is slightly below the initial 15% target, the model is significant enough to solve the problem as it provides a robust balance between precision and recall, ensuring that the company can identify a majority of at-risk customers accurately.

Section: Conclusion

Final Summary: The project successfully demonstrated that machine learning, specifically the **XGBoost** algorithm, can effectively predict customer churn in the telecommunications sector. Exploratory data analysis revealed that customers with **Month-to-month contracts** are the most likely to leave, providing a clear target for retention campaigns.



Business Recommendations:

1. **Targeted Incentives:** Proactively offer loyalty discounts to customers on monthly contracts.
2. **Model Deployment:** Implement the model in a production environment to monitor customer behavior in real-time.

8. References

1. **Vafeiadis, T., et al. (2015).** "A Comparison of Machine Learning Methods for Customer Churn Prediction." *Computer Engineering and Applications*.
2. **Chen, T., & Guestrin, C. (2016).** "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD*.
3. **He, H., & Garcia, E. A. (2009).** "Learning from Imbalanced Data." *IEEE Transactions on Knowledge and Data Engineering*.
4. **IBM Sample Data Sets.** "Telco Customer Churn." *Kaggle*.